System Architecture

OptiSight: Advanced Closed-Loop System Architecture

The system is designed as a **Hierarchical Vision-Reasoning Pipeline**, strategically decoupling immediate reaction from deep semantic analysis to ensure both safety and efficiency.

Layer 1: The Reflex Engine (Green Zone)

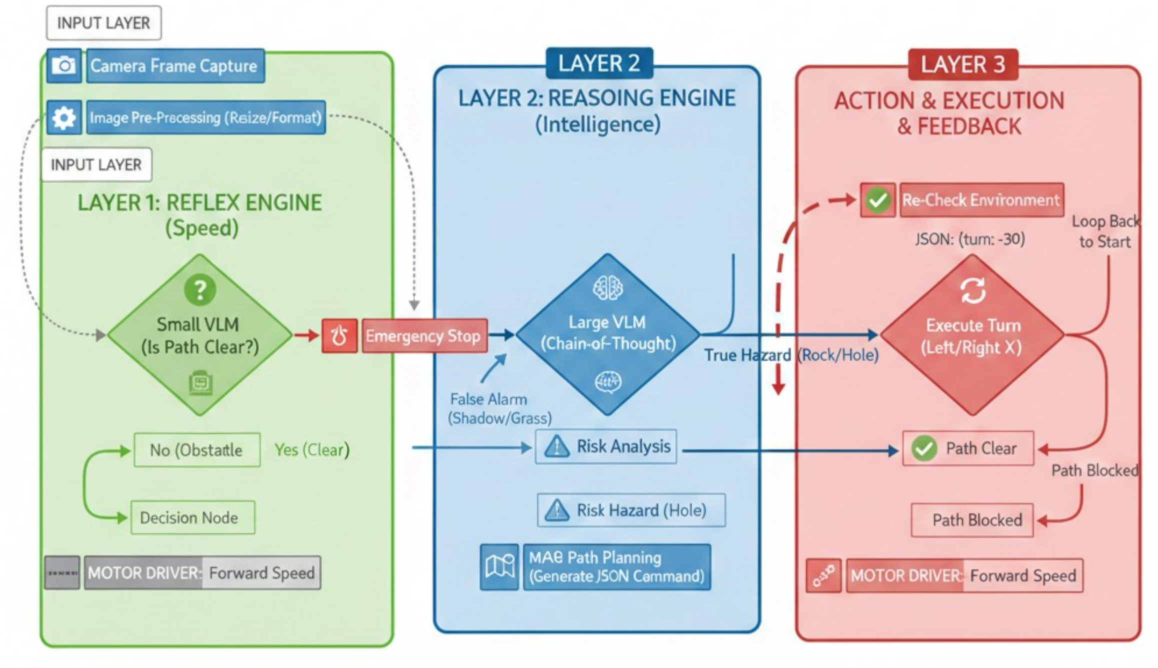
- **Focus:** High-Frequency Monitoring & Collision Avoidance.
- **Mechanism:** Utilizes a **Featherweight VLM** to perform binary path classification ("Clear" vs. "Blocked").
- **Operational Role:** Acting as the "Spinal Cord," it runs continuously at high FPS. If an anomaly is detected, it triggers an **Emergency Stop** and activates the next intelligence tier.

Layer 2: The Reasoning Engine (Blue Zone)

- **Focus:** Semantic Analysis & Risk Assessment.
- **Mechanism:** Employs a **Heavyweight VLM** using **Chain-of-Thought (CoT)** reasoning.
- **Operational Role:** Acting as the "Primary Brain," it filters false positives (e.g., shadows, soft grass). If a threat is confirmed (e.g., a deep hole or solid rock), it calculates a precise **Maneuver Plan** in JSON format.

Layer 3: Action & Feedback Loop (Red Zone)

- **Focus:** Execution & Safety Verification.
- **Mechanism:** Translates JSON commands into physical motor movement (e.g., specific turn degrees) and initiates a **Re-Check Phase**.
- **Operational Role:** Unlike linear systems, this is a **Closed-Loop**. After a maneuver, the system loops back to the Input Layer to verify the new path is safe before resuming forward speed.



Logic Flow

Key Technical Advantages

1. **Latency Optimization:** Computational resources are preserved by only activating the "Heavyweight Brain" during confirmed obstacle events.
2. **Autonomous Recovery:** Instead of a simple "Stop-and-Fail" logic, the system autonomously negotiates obstacles through intelligent maneuvers.
3. **Self-Verification:** The **Feedback Loop** ensures the robot never blindly executes a turn into a secondary hazard, providing a failsafe mechanism for complex terrains.